ORTHOGONALS TO A SHIFT-INVARIANT SUBCATEGORY ARE TRIANGULATED

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11964-11975, following the statement at 11954-11962.

Editorial note. The preservation under shifts, sums and summands is a definitional one-line reading, flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 11965 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Closure criterion for triangulated subcategories. lemma-triangulated-subcategory (same chapter, cited at ~11970). Reduces to the two-out-of-three closure of the orthogonal.
- Homological criterion for cone closure. lemma-pre-prepare-adjoint (carded L26232, cited at ~11973). Z lands in the orthogonal when X and Y do.

Source reference. Original definition 05RB: sources/stacks-derived/pinned-original-source.tex, source lines 1975–1975. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CQQ: sources/stacks-derived/pinned-original-source.tex, source lines 11900–11900. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QX: sources/stacks-derived/pinned-original-source.tex, source lines 1077–1077. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let $\mathcal{A} \subset \mathcal{D}$ be a full subcategory invariant under all shifts. Then both the right orthogonal $\mathcal{A}^\perp$ and the left orthogonal ${}^\perp\mathcal{A}$ of $\mathcal{A}$ are strictly full, saturated*[1]*, triangulated subcagories of $\mathcal{D}$.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

[1]Definition 05RB.

Proof. It is immediate from the definitions that the orthogonals are preserved under taking shifts, direct sums, and direct summands. Consider a distinguished triangle

$$X \to Y \to Z \to X[1]$$

of $\mathcal{D}$. By Lemma 05QX it suffices to show that if X and Y are in $\mathcal{A}^\perp$, then Z is in $\mathcal{A}^\perp$. This is immediate from Lemma 0CQQ. $\square$